

Article

Modeling Air Pollution in Metropolitan Lima: A Statistical and Artificial Neural Network Approach

Miguel Angel Solis Teran ¹, Felipe Leite Coelho da Silva ² , Elías A. Torres Armas ³, Natalí Carbo-Bustanza ⁴ and Javier Linkolk López-Gonzales ^{5,*} 

¹ Facultad de Ingeniería y Arquitectura, Universidad Peruana Unión, Lima 15468, Peru; mianso@upeu.edu.pe

² Department of Mathematics, Federal Rural University of Rio de Janeiro, Seropédica 23890-000, Brazil; felipeleite@ufrj.br

³ Instituto de Investigación de Estudios Estadísticos y Control de Calidad, Universidad Nacional Toribio Rodríguez de Mendoza, Chachapoyas 01001, Peru; elias.torres@untrm.edu.pe

⁴ E.P. Ingeniería Ambiental, Universidad Peruana Unión, Lima, Peru; natalicarbobustanza@gmail.com

⁵ Escuela de Posgrado, Universidad Peruana Unión, Lima 15468, Peru

* Correspondence: javierlinkolk@gmail.com

Abstract: Particulate matter is a mixture of fine dust and tiny droplets of liquid suspended in the air. PM₁₀ is a pollutant composed of particles smaller than 10 µm. These particles are harmful to the respiratory system. The air quality in the region and capital Lima in the Republic of Peru has been investigated in recent years. In this context, statistical analyses of PM₁₀ data with forecast models can contribute to planning actions that can improve air quality. The objective of this work is to perform a statistical analysis of the available PM₁₀ data and evaluate the quality of time series classical models and neural networks for short-term forecasting. This study demonstrates that classical time series models, particularly ARIMA and SSA, achieve lower average forecast errors than LSTM across stations SMP, CRB, and ATE. This finding suggests that for data with seasonal patterns and relatively short time series, traditional models may be more efficient and robust. Although neural networks have the potential to capture more complex relationships and long-term dependencies, their performance may be limited by hyperparameter settings and intrinsic data characteristics.

Keywords: neural network; modeling; statistical approach; air pollution; Lima



Academic Editors: Valerio Paolini and Francesco Petracchini

Received: 23 April 2025

Revised: 27 May 2025

Accepted: 4 June 2025

Published: 10 June 2025

Citation: Solis Teran, M.A.; Leite Coelho da Silva, F.; Torres Armas, E.A.; Carbo-Bustanza, N.; López-Gonzales, J.L. Modeling Air Pollution in Metropolitan Lima: A Statistical and Artificial Neural Network Approach. *Environments* **2025**, *12*, 196. <https://doi.org/10.3390/environments12060196>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Currently, the generation of pollutants in the air experiences a significant increase, predominantly in the form of gases and suspended particles of sizes dangerously small for human health. There is epidemiological evidence that particulate matter is associated with risks of cardiovascular and respiratory mortality [1,2]. According to Sánchez [3], it is crucial to understand this phenomenon from a spatiotemporal framework by studying elements that provide measurable data to explain the interaction between matter and energy in the environment. Similarly, Mahmud [4] suggests the importance of examining different elements within a dynamic field, which changes rapidly due to the variability of its spatial and temporal components, thus allowing objective, explainable, and understandable research in the environmental field.

According to [5,6], one of the methods that allows the modeling of suspended xenobiotics is the spatial and temporal analysis of the effects of PM₁₀, as well as the influence of meteorological variables such as temperature and wind on air quality. In [7], authors highlight that the understanding of these particles, due to their size, is worrying, especially due

to their levels of permanence and invasion in the respiratory tract. In this context, ref. [8] makes favorable predictions through singular spectral analysis. This approach is considered a reliable stochastic process to draw a spatiotemporal visualization of PM_{10} , a pollutant with a great impact on exposed populations.

Singular spectral analysis (SSA) has proven to be an effective tool in the prediction of air pollutants such as PM_{10} due to its ability to decompose complex time series into simpler components, thereby facilitating the identification of underlying patterns and trends in air quality data. According to study [9], the combination of SSA with machine learning techniques significantly improves the accuracy in prediction of airborne pollutants. Furthermore, the integrated SSA-ARIMA approach described in [10] enables multi-day forecasts with high confidence, applicable also to PM_{10} . For its part, study [11] presents hybrid deep learning models that, combined with SSA, offer high performance in the long-term prediction of pollutants such as PM_{10} and $PM_{2.5}$, highlighting the value of advanced techniques in the evaluation and control of air quality.

Two studies, that of Bodor et al. (2020) in Transylvania [12] and that of Shikhovtsev et al. (2023) in the South Baikal Region [13], coincide in highlighting the seasonal and spatial variability of PM_{10} concentrations, demonstrating the influence of both local climatic conditions and long-range transport on the dynamics of this atmospheric pollutant. In addition, the studied attempt to take the sensory perception of pollutants towards a current, agile, and dynamic quantification system, capable of being modeled and understood through statistical processes. On the other hand, ref. [14] states that there are frequent variability measures within the spatial components, subject to certain meteorological factors that can generate white noise. However, ref. [15] establishes that the time series comprise a dynamic system, defined as a linear combination of various oscillators, which through series projection in principal components developed in the time-frequency domain allows establishing dominant oscillation models and associating them with various events compromised with PM_{10} particulates. Ref. [16] mentions that singular spectral analysis (SSA) is a fundamental part of the probabilistic prediction of the energy process, autoregressive models to predict modular behaviors of matter. For [17,18], the usefulness of the SSA method accompanied by others that helped to decompose the data improved the spectral analysis of PM_{10} emission, considering its proportions and effects on the environment.

Finally, it is worth pointing out that few studies have been carried out in Peru related to suspended particulate pollutants. The last study related to pollutants smaller than 10 microns in [19,20] analyzed the spatial and temporal distribution of PM_{10} concentration in Metropolitan Lima in the period 2015–2017 with results above quality standards; however, no specific spectral studies have been carried out in order to trace sufficient frequencies to understand pollution within atmospheric patterns. In [21], the authors note that concentrations exhibit non-linear behavior and fluctuate strongly on space-time scales, which allows to visualize and describe widespread pollution and trace its permanence. In that sense, the objective of this research is to model and forecast the concentration of PM_{10} per hour based on artificial neural networks and traditional models.

2. Methodology

In this study, PM_{10} data from Peru were used. The National Meteorological and Hydrological Service uses particle samplers to collect PM_{10} . These devices are part of the Automatic Air Quality Monitoring Network that SENAMHI operates to monitor air quality. The samplers are semi-manual and use 47 mm filters to capture airborne particles. This analysis considered five weather stations located in the region of the province of Lima. The study area comprises the capital of the Republic of Peru, the province of Lima

(Figure 1). Lima is located at 77° W and 12° S of the South American continent. The data used covers the period of 2017–2018.

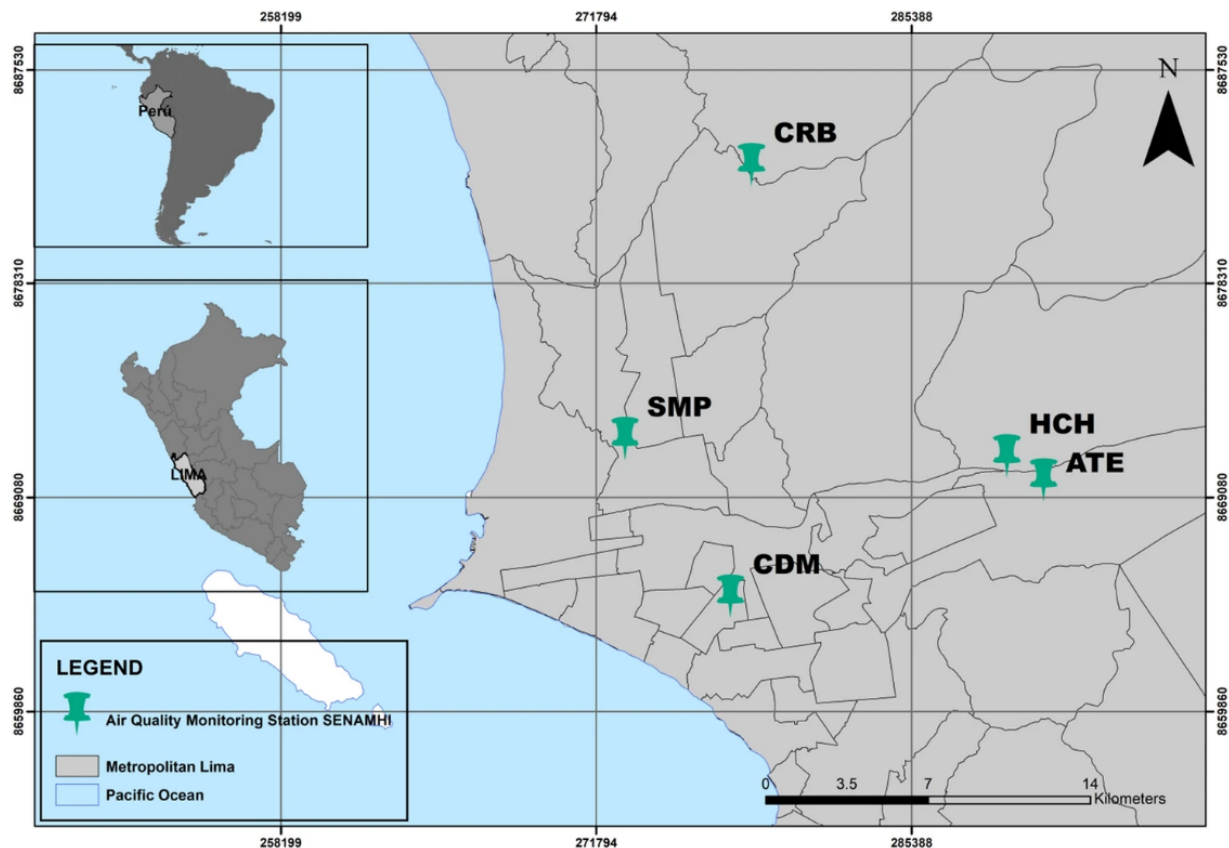


Figure 1. Map with study area and locations of the Lima air quality monitoring stations: Ate (ATE), Campo de Marte (CDM), Carabayillo (CRB), Huachipa (HCH), and San Martín de Porres (SMP). Scale 1 cm/7 km.

To evaluate the prediction accuracy of the models, the PM_{10} time series was divided into 10 training and test sets. From the first defined training set, a forecast of seven days ahead was carried out. Then, the training set received one more observation, and again the forecast was performed seven days ahead. Following this methodology, the steps mentioned above were carried out 10 times. In this work, the classical time series models were used, such as the Box–Jenkins and exponential smoothing models. Also, autoregressive neural network models (NNARs), multilayer perceptrons (MLPs), and long short-term memory (LSTM) models were used. Furthermore, the naive models, TBATS, dynamic linear model (DLM), and singular spectrum analysis (SSA) were adopted.

2.1. Box–Jenkins Models

The Box–Jenkins [22] methodology is widely used in analyzing parametric time series models. This methodology includes fitting integrated autoregressive moving average models, $ARIMA(p, d, q)$, to a data set using the autocorrelation functions between observations.

The model used was seasonal ARIMA (SARIMA), which incorporated the seasonality component into the data. The structure of the seasonal ARIMA model of order $(p, d, q) \times (P, D, Q)_s$ is given by

$$\phi(B)\Phi(B^s)\Delta^d\Delta_s^D Z_t = \theta(B)\Theta(B^s)a_t \quad (1)$$

where a_t is white noise; $\phi(B)$ is the autoregressive operator of order p ; $\theta(B)$ is the moving average operator of order q ; $\Phi(B^s)$ is the seasonal autoregressive operator of order P ; $\Theta(B^s)$ is the seasonal moving average operator of order Q ; Δ^d is the simple difference operator;

Δ_s^D is the seasonal difference operator; s is the number of observations per year (period). The Box–Jenkins model was obtained using the algorithm proposed by Hyndman and Khandakar [23].

2.2. Exponential Smoothing Method

The exponential smoothing method was proposed in the 1950s in [24–26]. In this method, forecasts are generated from a weighted average of past observations, with the weight of observations decreasing exponentially as the observations age. Hyndman et al. [27] proposed a classification that depends on the time series' error, trend, and seasonality. The error can be additive (A) or multiplicative (M), the trend can be additive (A), additive with damping (A_d) or, in the case of non-existence, there can be none (N); in turn, seasonality can be additive (A), multiplicative (M), or there can be none (N). Therefore, each exponential smoothing model used in this work can be classified as ETS (Error, Trend, Seasonality). Furthermore, the ETS algorithm proposed by Hyndman [28] was used to adjust the exponential smoothing models.

2.3. Neural Network Autoregression

The Neural Network Autoregression (NNAR) model constitutes a linear generalization of classic autoregression models, in which the relationship between the calculated values of a time series and its current value is modeled using a neural network feedforward from a hidden layer. This paper uses the algorithm proposed in [23]. Formally, a NNAR(p, k) model can be expressed as

$$y_t = f(y_{t-1}, y_{t-2}, \dots, y_{t-p}) + \varepsilon_t \quad (2)$$

where $f(\cdot)$ represents a non-linear function approximated by a neuronal network with k neurons in the hidden layer and $y \varepsilon_t$ is an error term with zero mean and constant variance. The neural network is trained using optimization algorithms such as backpropagation with gradient descent, minimizing loss function typically based on the average quadratic error [27,29]. The NNAR architecture is particularly useful in contexts where temporal dynamics present linear, seasonal, or wide-range behaviors that are not captured efficiently by linear models such as ARIMA. Furthermore, its implementation maintains the traditional autoregressive structure, which facilitates comparison and interpretation within the time series modeling framework [30].

2.4. Multi-Layer Perceptron

The MLP model or the feedforward neural network is a mathematical function mapping a sort of input values to output values. According to [31], the main objective of a feedforward neural network is to approximate any function f^* defining a mapping $Y = f(x; \theta)$ and learn the value of the parameter θ that makes the better function approximation. This consists of at least three node covers: an input layer, one or more hidden covers, and an output layer. Each node (except those at the input layer) applies a linear activation function, which allows the model to capture complex and non-linear relationships between the input and output variables. Formally, the MLP output is defined as

$$\hat{y} = f^{(L)} \left(W^{(L)} f^{(L-1)} \left(\dots f^{(1)} \left(W^{(1)} x + b^{(1)} \right) + \dots \right) + b^{(L)} \right) \quad (3)$$

where L is the number of covers, $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the l cover, and $f^{(l)}$ is the corresponding activation function. Commonly employed activation functions include ReLU, sigmoid, and tanh [31,32]. MLP training is carried out using a backpropagation algorithm in combination with optimization techniques such as stochastic gradient

descent or more sophisticated variants such as Adam. Due to their flexibility and universal approximation capacity [33], MLPs are used as fundamental blocks in more complex architectures of deep learning.

2.5. Long Short-Term Memory

Long short-term memory (LSTM) networks are a variant of recurrent neural networks (RNNs) that effectively solve the problem of gradient fading when modeling long-term dependencies in data sequences [34]. Their architecture incorporates a memory cell and computer mechanisms to regulate the flow of information. The behavior of an LSTM unit can be expressed in compact form as

$$\begin{bmatrix} f_t \\ i_t \\ o_t \\ \tilde{C}_t \end{bmatrix} = \begin{bmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{bmatrix} \left(W \begin{bmatrix} h_{t-1} \\ x_t \end{bmatrix} + b \right)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad h_t = o_t \odot \tanh(C_t)$$

where x_t is the input at time t , h_{t-1} is the previous hidden state, C_t is the state of the cell, σ denotes the sigmoid function, and $\odot$ represents the product element by element. LSTMs are widely used in time series prediction tasks due to their ability to capture complex patterns and wide-ranging dependencies [35,36].

2.6. Dynamic Linear Model

Dynamic Linear Models (DLMs) form a class of Bayesian state-space models used for time series with structures that evolve over time [37]. Their formulation is based on two normal conditional distributions: $y_t \mid \theta_t \sim \mathcal{N}(F_t^\top \theta_t, V_t)$ for the observation model and $\theta_t \mid \theta_{t-1} \sim \mathcal{N}(G_t \theta_{t-1}, W_t)$ for the evolution model. Here, y_t is the observation at t time, θ_t is the latent state vector, F_t is the regression vector, G_t is the transition matrix, and V_t, W_t are the variance matrices associated with observation and evolution errors, respectively. This structure allows dynamic capture of changes in trend, seasonality, and other unobserved components, with estimation based on the Kalman filter and sequential Bayesian methods [38].

3. Results

In this section, an exploratory analysis is presented for PM₁₀ data from meteorological stations in Peru. Then, the time series models are used to obtain short-term forecasting of PM₁₀.

3.1. Exploratory Analysis

Figure 2 shows the PM₁₀ concentration trajectory of the monitoring station in Peru. The HCH station presented the highest PM₁₀ concentration followed by the ATE station. On the other hand, the CDM station had the lowest daily PM₁₀ concentration rates (Figure 3). The highest daily PM₁₀ concentration value occurred in April 2018 at the HCH meteorological station. Figure 3 shows the boxplot graph for the hourly and daily PM₁₀ data. It can be seen in this figure that both data sets have discrepant observations. Figure 4 shows the daily time series of PM₁₀ and the boundaries of the boxplot. This figure shows that the outliers occur in different months of the year.

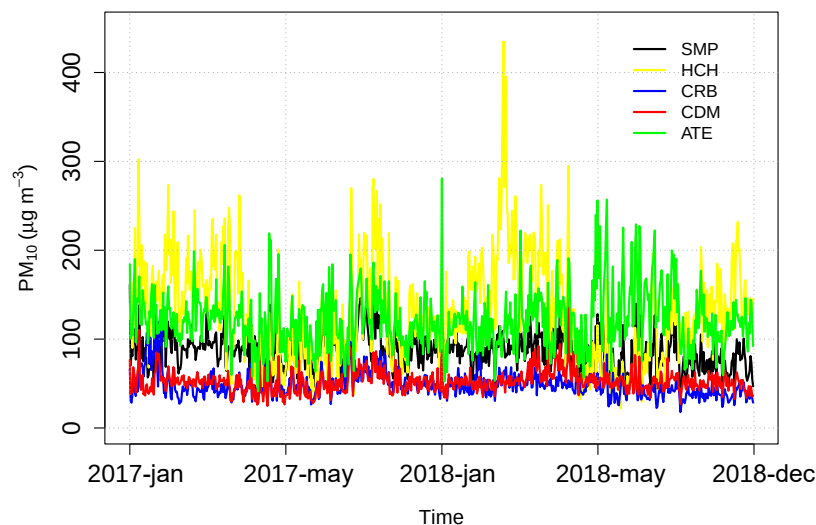


Figure 2. Average daily concentration of PM_{10} between 2017 and 2018 for monitoring station in Peru.

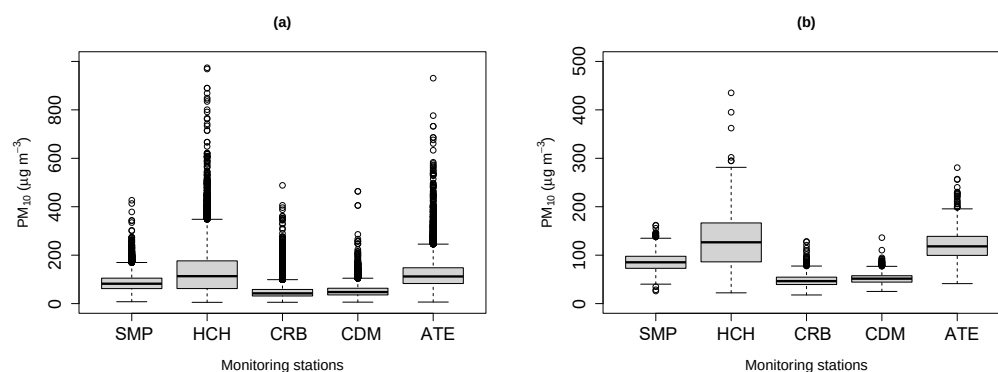


Figure 3. Boxplots of hourly (a) and daily (b) PM_{10} concentrations at five monitoring stations located in Peru. This figure presents a comparative analysis of PM_{10} levels recorded at stations SMP, HCH, CRB, CDM, and ATE. In panel (a), the boxplots reflect the distribution of hourly concentrations, revealing high temporal variability and the presence of numerous outliers, especially at HCH and SMP, suggesting specific episodes of intense pollution. In panel (b), the data aggregated at the daily level show a more contained dispersion, although high maximum values are still observed at some stations, again highlighting HCH and SMP as critical areas. This type of visualization makes it possible to identify differences in pollution patterns between stations, which is key to establishing local air quality management strategies.

Table 1 provides the main statistical measures of daily PM_{10} concentration for meteorological stations considered in this study. It can be seen in this table that the HCH station has the highest average concentration of PM_{10} and also greater variability, and the ATE station has the lowest average PM_{10} . Hourly temperature and PM_{10} data at each station analyzed did not present a correlation (Figure 5). The HCH station presented the highest correlation between stations, being approximately 0.4, still a low value. Figure 6a presents the spatial correlation between weather stations considered in Peru. This figure shows that the highest correlation (0.63) occurred between the SMP and ATE stations. On the other hand, the spatial correlation between monitoring stations for the temperature variable is greater than 0.70 (Figure 6b). Therefore, there is a low spatial correlation between some air quality monitoring stations for variable PM_{10} compared to the temperature variable.

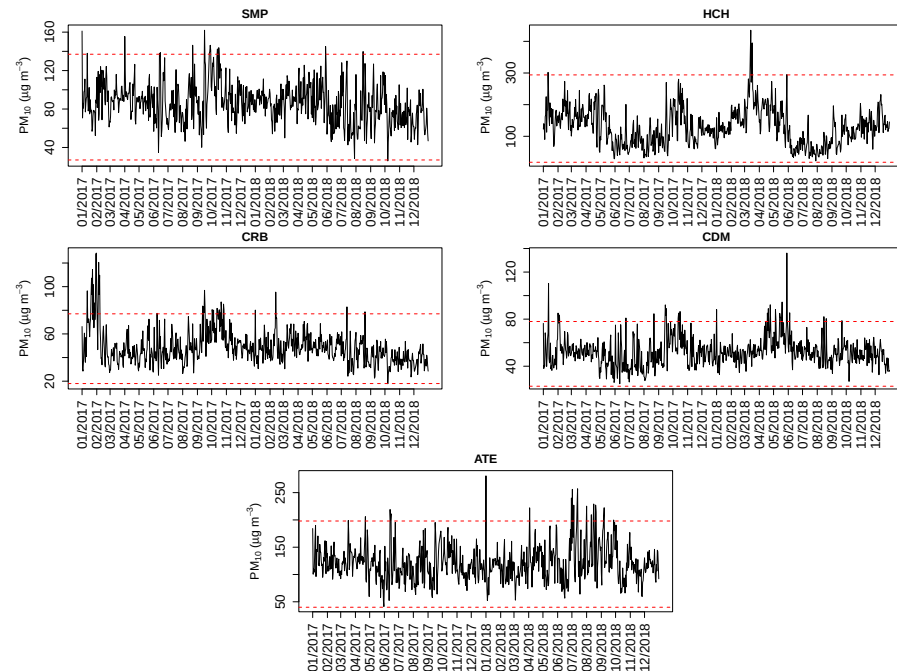


Figure 4. Daily PM_{10} time series with boxplot boundaries.

Table 1. Statistical measures for daily PM_{10} ($\mu\text{g} \cdot \text{m}^{-3}$) concentration from weather stations.

Measures	SMP	HCH	CRB	CDM	ATE
Minimum	26.01	22.22	17.84	25.12	41.26
1st Qu.	72.89	86.21	39.38	44.35	99.59
Median	85.29	126.61	46.40	51.46	118.19
Mean	86.05	130.03	48.69	52.30	121.56
3rd Qu.	97.69	166.56	54.63	57.51	138.72
Maximum	161.97	435.15	128.44	136.16	280.72
Standard deviation	20.88	58.63	14.50	12.25	33.29

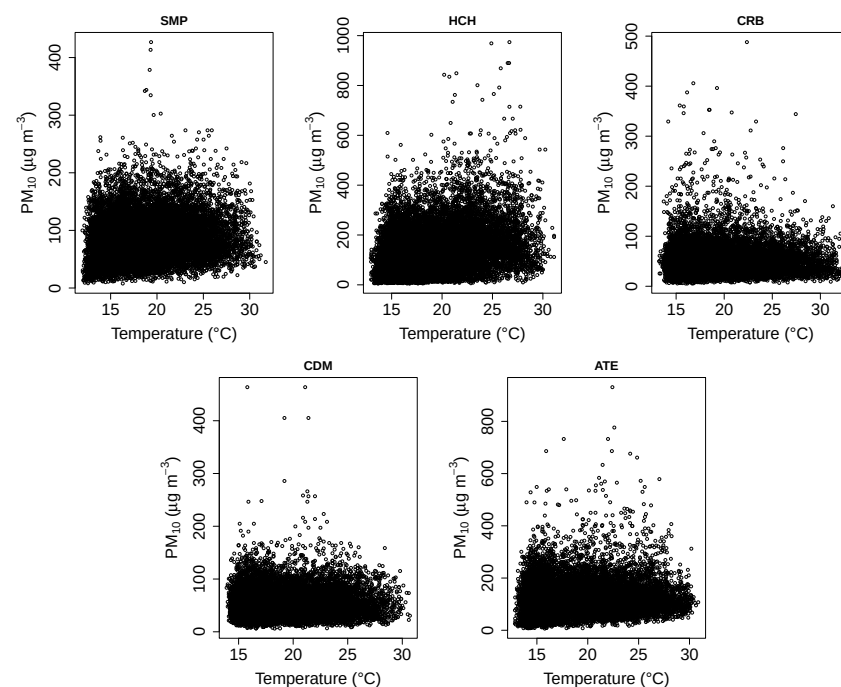


Figure 5. Scatterplot between temperature and PM_{10} at each meteorological station (SMP, HCH, CRB, CDM, ATE) in Peru.

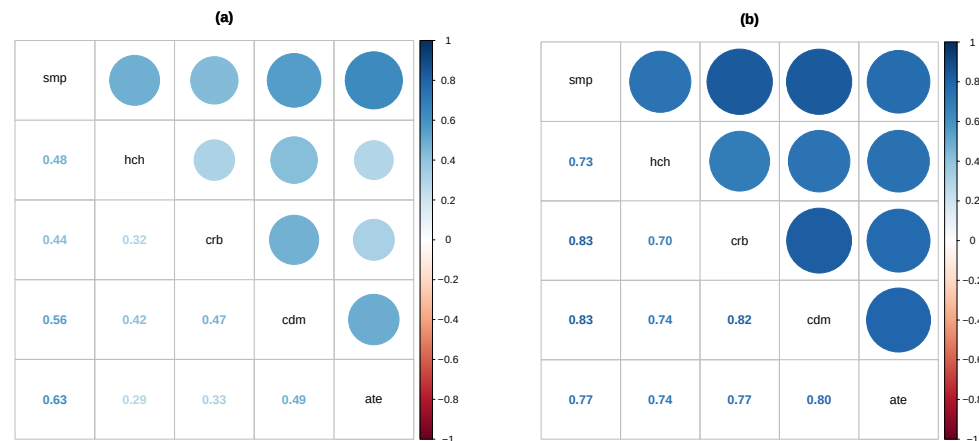


Figure 6. Spatial correlation between weather stations for PM₁₀ (a) and temperature (b) in Peru.

3.2. Model Performance Evaluation for PM₁₀ Forecasting in Lima

Table 2 presents the RMSE and sMAPE accuracy measures for the forecast models in each test set and the average. Based on the average values of the RMSE and sMAPE metrics, the Box–Jenkins model presents the best prediction accuracy for PM₁₀ data of SMP station. In this station, the models present a symmetric mean absolute percentage error measure lower than 10%, except the DLM model. In Table 3, the LSTM model presents the best predictive capacity for the data from the HCH station. For the CRB station data, the TBATS model provides better prediction results (see Table 4). Table 5 shows that the SSA model has the best prediction results for the ATE station data.

Table 2. Forecasting performance of the nine models applied to the PM₁₀ ($\mu\text{g} \cdot \text{m}^{-3}$) data of the SMP station via RMSE and sMAPE metrics.

Metric	Sets	ETS	ARIMA	TBATS	NNAR	MLP	DLM	SSA	NAïVE	LSTM
RMSE	1	8.073	12.131	9.991	8.804	10.914	39.730	8.081	15.614	9.683
	2	11.862	13.691	12.284	13.873	13.460	41.701	13.232	12.470	12.661
	3	15.951	15.123	14.930	19.771	17.941	13.320	18.412	15.572	18.391
	4	18.141	16.362	16.954	22.173	20.874	13.701	21.701	17.232	20.704
	5	20.410	15.962	17.771	25.403	23.171	38.762	23.761	22.962	24.494
	6	24.390	15.511	18.872	28.314	26.591	82.103	24.872	37.531	25.174
	7	23.804	15.341	18.063	26.942	24.394	15.911	25.822	21.823	21.444
	8	21.773	15.001	17.473	24.881	23.640	27.771	26.223	15.383	24.580
	9	16.913	12.884	14.763	18.961	20.190	66.444	25.323	10.751	24.394
	10	13.833	14.052	15.020	14.691	20.084	48.243	20.801	14.054	23.583
	Average	17.514	14.603	15.611	20.383	20.120	38.774	20.823	18.343	20.511
sMAPE (%)	1	1.871	2.801	2.030	2.193	2.954	9.184	1.853	4.150	2.621
	2	2.714	3.643	2.951	3.094	3.091	17.170	3.113	2.981	2.964
	3	4.130	4.213	3.981	4.950	4.444	3.691	4.633	4.080	4.701
	4	5.240	5.041	5.034	6.260	5.691	4.194	5.833	5.121	5.611
	5	6.234	5.081	5.543	7.573	6.901	10.322	7.032	6.890	7.292
	6	7.560	5.022	6.091	8.863	8.363	18.440	7.621	11.152	7.802
	7	7.484	5.044	5.731	8.440	7.603	4.954	8.011	6.853	6.734
	8	6.921	4.982	5.652	8.011	7.580	10.091	8.001	5.092	7.804
	9	5.564	4.423	4.981	6.232	6.512	33.884	7.920	3.071	7.743
	10	4.732	4.554	4.910	4.891	6.274	30.032	6.742	4.474	7.473
	Average	5.242	4.480	4.693	6.054	5.942	14.190	6.071	5.382	6.074

Table 3. Forecasting performance of the nine models applied to the PM₁₀ ($\mu\text{g} \cdot \text{m}^{-3}$) data of the HCH station via RMSE and sMAPE metrics.

Metric	Sets	ETS	ARIMA	TBATS	NNAR	MLP	DLM	SSA	NAïVE	LSTM
RMSE	1	75.224	41.614	37.843	44.831	44.084	41.332	47.854	56.384	21.922
	2	61.504	42.791	39.084	46.531	40.504	62.163	58.983	36.132	21.102
	3	43.873	37.374	34.541	36.793	30.843	82.413	57.581	22.631	20.662
	4	19.683	23.664	20.934	36.603	15.354	167.442	43.674	33.824	15.742
	5	24.712	26.954	25.173	41.804	17.403	18.761	31.742	16.421	20.302
	6	24.451	22.964	22.123	50.734	16.931	17.262	18.214	17.161	16.104
	7	42.721	30.093	28.904	49.704	21.153	98.882	14.402	32.391	17.152
	8	53.264	33.733	32.884	65.842	29.553	77.753	11.163	39.013	17.994
	9	26.482	22.543	21.043	23.404	17.013	107.383	9.194	12.692	8.743
	10	25.843	24.072	23.392	36.624	18.894	11.324	13.321	12.403	11.953
	Average	39.773	30.582	28.594	43.281	25.173	68.472	30.613	27.902	17.161
sMAPE (%)	1	9.963	5.761	5.214	6.123	5.942	5.404	6.564	7.741	3.102
	2	8.724	6.183	5.612	6.621	5.701	11.803	8.443	5.101	2.924
	3	6.453	5.584	5.121	5.462	4.513	19.271	8.374	3.583	2.944
	4	2.963	3.103	2.943	4.663	2.312	39.583	6.472	5.754	2.383
	5	3.874	3.932	3.732	5.614	2.912	2.981	4.721	2.334	2.591
	6	4.071	3.393	3.521	6.902	2.521	2.553	2.613	2.423	2.324
	7	6.451	4.553	4.513	7.541	3.394	12.284	2.354	5.034	2.302
	8	8.374	5.542	5.451	9.841	4.931	11.164	1.471	6.463	2.953
	9	3.942	3.604	3.392	3.463	2.861	31.184	1.194	1.982	1.371
	10	4.021	3.874	3.731	5.381	2.984	1.771	1.751	2.183	1.532
	Average	5.881	4.553	4.324	6.161	3.814	13.804	4.393	4.261	2.442

Table 4. Forecasting performance of the nine models applied to the PM₁₀ ($\mu\text{g} \cdot \text{m}^{-3}$) data of the CRB station via RMSE and sMAPE metrics.

		ETS	ARIMA	TBATS	NNAR	MLP	DLM	SSA	NAïVE	LSTM
RMSE	1	5.614	5.294	5.302	7.401	7.904	20.104	5.671	9.631	7.751
	2	6.302	6.642	6.504	7.814	7.574	40.282	6.641	8.911	9.224
	3	7.121	7.072	6.884	9.431	9.102	8.134	7.754	7.102	11.162
	4	7.284	7.042	6.841	10.421	9.344	12.491	7.754	7.521	12.614
	5	7.744	7.262	6.882	11.304	10.024	12.194	8.292	8.214	12.284
	6	10.312	8.872	8.574	13.274	12.654	38.324	8.861	16.021	14.074
	7	9.571	7.962	6.431	11.934	11.464	8.861	9.194	8.112	13.574
	8	9.181	8.192	7.854	12.114	11.764	6.972	9.504	8.042	15.804
	9	4.332	4.182	3.504	6.964	6.874	45.434	8.214	8.401	11.964
	10	4.384	5.424	5.094	9.294	8.764	11.741	6.934	5.542	12.574
	Average	7.181	6.792	6.384	9.994	9.544	20.454	7.884	8.754	12.104
SMAPE (%)	1	2.854	2.134	2.114	3.864	4.024	9.374	2.661	4.741	3.961
	2	2.454	3.444	3.214	3.314	3.094	35.394	2.991	5.264	4.484
	3	3.404	3.694	3.604	4.494	4.154	4.354	3.584	3.884	5.454
	4	3.534	3.534	3.484	5.284	4.624	5.944	3.654	3.484	6.284
	5	3.954	3.823	3.654	6.164	5.294	6.402	4.191	4.251	6.463
	6	5.752	5.042	4.854	7.543	7.171	16.314	4.754	8.874	7.874
	7	5.334	4.322	3.604	6.704	6.421	5.542	4.953	4.351	7.653
	8	5.261	4.641	4.314	7.132	6.954	3.851	5.054	4.501	8.924
	9	2.444	2.222	1.974	3.941	3.844	39.854	4.544	5.741	6.891
	10	2.604	3.062	2.904	5.002	4.391	9.141	3.964	3.291	7.221
	Average	3.764	3.591	3.374	5.341	4.994	13.614	4.031	4.841	6.523

Table 5. Forecasting performance of the nine models applied to the PM₁₀ ($\mu\text{g} \cdot \text{m}^{-3}$) data of the ATE station via RMSE and sMAPE metrics.

		ETS	ARIMA	TBATS	NNAR	MLP	DLM	SSA	NAïVE	LSTM
RMSE	1	13.831	13.782	14.802	12.973	12.321	15.751	15.474	12.301	12.622
	2	12.881	13.231	15.252	12.763	12.612	31.471	15.642	14.322	12.992
	3	19.191	17.412	16.023	18.601	16.634	29.031	15.974	17.692	16.041
	4	25.511	22.161	19.132	23.151	21.721	32.961	17.302	23.602	19.822
	5	20.733	18.922	18.991	19.453	20.123	60.593	16.252	20.951	18.602
	6	21.412	19.352	18.251	19.771	19.973	25.042	15.791	19.183	21.363
	7	22.854	22.073	21.512	22.803	21.896	27.723	20.553	22.222	23.534
	8	26.202	23.584	22.763	23.602	26.193	101.282	21.364	37.993	21.614
	9	23.114	20.401	20.432	19.653	21.254	62.064	20.903	21.144	21.902
	10	20.712	19.264	19.779	19.883	19.585	107.774	21.863	31.865	21.674
	Average	20.643	19.024	18.693	19.265	19.235	49.375	18.114	22.124	19.013
SMAPE (%)	1	2.403	2.262	2.324	2.083	1.883	2.422	2.172	1.873	1.952
	2	2.135	2.064	2.393	1.965	2.013	6.143	2.197	2.164	1.902
	3	3.342	3.042	2.553	3.086	2.713	4.674	2.485	3.052	2.594
	4	4.802	4.193	3.373	4.314	4.025	5.893	3.091	4.506	3.484
	5	3.403	3.104	3.422	3.213	3.432	17.782	2.903	4.113	3.004
	6	3.812	3.401	3.104	3.564	3.562	4.613	2.682	3.324	3.903
	7	4.362	4.123	3.873	4.214	4.083	5.162	3.613	4.146	4.553
	8	4.864	4.521	4.424	4.613	5.013	15.074	3.736	6.443	4.071
	9	4.423	3.723	3.792	3.674	4.093	14.294	3.885	3.944	4.092
	10	3.774	3.602	3.673	3.724	3.564	33.742	4.013	5.803	3.963
	Average	3.733	3.404	3.295	3.443	3.441	10.983	3.076	3.933	3.352

Finally, it is worth pointing out that few studies have been carried out in Peru related to suspended particulate pollutants. The last study related to pollutants smaller than 10 microns by Espinoza [19] analyzed the spatial and temporal distribution of PM₁₀ concentration in Metropolitan Lima in the period of 2015–2017 with results above quality standards; however, no specific spectral studies have been carried out in order to trace sufficient frequencies to understand pollution within atmospheric patterns. The superior performance of classical models such as ARIMA compared to LSTM neural networks raises a relevant discussion about the suitability of modeling tools to the characteristics of the data. Although neural networks are designed to capture nonlinear relationships and long-term dependencies, their effectiveness depends on appropriate hyperparameter settings, a large amount of training data, and complex patterns in the time series. In this study, the available data show seasonal patterns and more predictable dependencies, characteristics that classical models efficiently handle. This highlights that the simplicity and adaptability of classical models can be advantages in scenarios with data limitations or well-defined patterns.

Analysis of PM₁₀ concentrations at different monitoring stations in Metropolitan Lima reveals significant spatial heterogeneity, with Huachipa (HCH) recording the highest levels. This finding may be related to specific emission sources, such as industries and heavy traffic, underlining the need for targeted local policies. Furthermore, the observed temporal fluctuations, influenced by factors such as seasons and meteorological changes, indicate the importance of considering spatiotemporal dynamics when planning interventions.

The results of this study partially align with international research highlighting the usefulness of advanced tools such as SSA and hybrid models to predict pollutants. However, the lower effectiveness of neural networks in this case highlights the importance of adapting models to local particularities. This includes not only adjusting the techniques used, but also expanding the database to improve predictive capacity. Likewise, a more robust integration of factors such as environmental policies, urban growth, and human behavior could enrich the models and improve mitigation strategies. In this sense, the study raises the need

for an interdisciplinary approach to effectively address the challenges of air pollution in Metropolitan Lima and other similar cities.

3.3. Limitations and Future Work

Despite the progress made in modeling air pollution in Metropolitan Lima using statistical approaches and artificial neural networks, the study has some limitations that should be considered. First, the study used a small number of monitoring stations, which limits the robustness of the models and the spatial representativeness of the results. Furthermore, the restricted geographic coverage prevents a broader assessment of pollutant behavior throughout the metropolitan area. Finally, the exclusion of relevant meteorological variables such as humidity and precipitation could have affected the accuracy of the estimates, given that these conditions directly influence the dispersion and concentration of air pollutants. In order to improve the accuracy and applicability of air pollution models in Metropolitan Lima, it is recommended to expand the network of monitoring stations to achieve greater spatial representativeness and better geographic coverage. This would allow for more accurate capture of local variations in pollution levels. It is also suggested that additional meteorological variables, such as relative humidity, precipitation, and atmospheric pressure, be incorporated, as these can significantly influence pollutant dynamics. The inclusion of these factors could enhance the predictive capacity of the models, especially in approaches based on artificial neural networks. Finally, the possibility of applying deep learning techniques with more complex architectures and larger data sets is proposed to explore deeper nonlinear patterns and improve the generalizability of the results.

4. Conclusions

This study demonstrates that classical time series models, particularly ARIMA and SSA, achieve lower average forecast errors than LSTM across stations SMP, CRB, and ATE. This finding suggests that for data with seasonal patterns and relatively short time series, traditional models may be more efficient and robust. Although neural networks have the potential to capture more complex relationships and long-term dependencies, their performance may be limited by hyperparameter settings and intrinsic data characteristics.

The high PM₁₀ concentrations at HCH station may be driven by nearby industrial sources and dense vehicular traffic, warranting targeted monitoring and policy intervention. These results have important implications for public health, since PM₁₀ particles are directly linked to respiratory and cardiovascular diseases. In addition, the levels recorded at several stations exceed international air quality standards, highlighting the urgent need to implement pollution mitigation policies in Metropolitan Lima. The use of advanced tools such as singular spectral analysis (SSA) and neural networks, together with classical models, provides valuable insight for air pollutant prediction. However, the results underline the importance of adapting these approaches to local conditions and increasing temporal and spatial resolution of measurements to enhance model reliability. Future research could explore hybrid models that combine the best of classical and modern approaches, as well as incorporate exogenous variables such as wind speed, relative humidity, precipitation, and traffic intensity to capture pollutant transport and accumulation dynamics. Furthermore, the findings may extend to other scenarios, such as those addressed in [39–42]. This will strengthen environmental management strategies and reduce the impact of pollution on public health.

Author Contributions: Conceptualization, M.A.S.T., F.L.C.d.S., E.A.T.A., N.C.-B. and J.L.L.-G.; methodology, M.A.S.T., F.L.C.d.S., E.A.T.A., N.C.-B. and J.L.L.-G.; software, M.A.S.T., F.L.C.d.S. and E.A.T.A.; validation, N.C.-B.; formal analysis, F.L.C.d.S. and J.L.L.-G.; investigation, M.A.S.T.;

resources, F.L.C.d.S. and J.L.L.-G.; data curation, F.L.C.d.S. and J.L.L.-G.; writing—original draft preparation, M.A.S.T., F.L.C.d.S., E.A.T.A., N.C.-B. and J.L.L.-G.; writing—review and editing, M.A.S.T., F.L.C.d.S., E.A.T.A., N.C.-B. and J.L.L.-G.; visualization, N.C.-B. and J.L.L.-G.; supervision, N.C.-B. and J.L.L.-G.; project administration, N.C.-B. and J.L.L.-G.; funding acquisition, E.A.T.A. and J.L.L.-G. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data sets are available in the repository <https://www.senamhi.gob.pe/site/descarga-datos/>, accessed on 15 January 2025.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Hoek, G.; Krishnan, R.M.; Beelen, R.; Peters, A.; Ostro, B.; Brunekreef, B.; Kaufman, J.D. Long-term air pollution exposure and cardio-respiratory mortality: A review. *Environ. Health* **2013**, *12*, 43. [CrossRef] [PubMed]
2. Chen, J.; Hoek, G. Long-term exposure to PM and all-cause and cause-specific mortality: A systematic review and meta-analysis. *Environ. Int.* **2020**, *143*, 105974. [CrossRef] [PubMed]
3. Sánchez-Balseca, J.; Pérez-Foguet, A. Spatio-temporal air pollution modelling using a compositional approach. *Heliyon* **2020**, *6*, e04794. [CrossRef]
4. Mahmud, H.; Shobnom, K.; Ali, M.R.; Muntakim, N.; Kulsum, U.; Baroi, D.S.; Ahmed, Z.; Rahman, M.M.; Hassan, M.Z. Micro-environmental dynamics of particulate (PM_{2.5} and PM₁₀) air pollution in Rajshahi City: A spatiotemporal analysis. *Manag. Environ. Qual. Int. J.* **2024**, *35*, 1773–1797. [CrossRef]
5. Akdi, Y.; Gölveren, E.; Ünlü, K.D.; Yücel, M.E. Modeling and forecasting of monthly PM_{2.5} emission of Paris by periodogram-based time series methodology. *Environ. Monit. Assess.* **2021**, *193*, 622. [CrossRef]
6. Leng, S.; Gao, X.; Pei, T.; Zhang, G.; Chen, L.; Chen, X.; He, C.; He, D.; Li, X.; Lin, C.; et al. Tempo-Spatial Processes and Modelling of Environmental Pollutants. In *The Geographical Sciences During 1986–2015: From the Classics to the Frontiers*; Springer: Singapore, 2017; pp. 367–390.
7. Thompson, J.E. Airborne particulate matter: Human exposure and health effects. *J. Occup. Environ. Med.* **2018**, *60*, 392–423. [CrossRef]
8. Cekim, H.O. Forecasting PM10 concentrations using time series models: A case of the most polluted cities in Turkey. *Environ. Sci. Pollut. Res.* **2020**, *27*, 25612–25624. [CrossRef] [PubMed]
9. Performance Analysis of Machine Learning Singular Spectrum Analysis for Forecasting Air Contamination. In Proceedings of the 2023 International Conference on System, Computation, Automation and Networking (ICSCAN), Puducherry, India, 17–18 November 2023. [CrossRef]
10. Kumar, U. An integrated SSA-ARIMA approach to make multiple day ahead forecasts for the daily maximum ambient O₃ concentration. *Aerosol Air Qual. Res.* **2015**, *15*, 208–219. [CrossRef]
11. Cai, J.; Gu, C.; Fang, K.; Wang, L.; Lv, M. Long-term PM_{2.5} Concentration Prediction Using Hybrid Deep Learning Model. In Proceedings of the 2023 4th International Conference on Computers and Artificial Intelligence Technology (CAIT), Macau, Macao, 13–15 December 2023. [CrossRef]
12. Bodor, Z.; Bodor, K.; Keresztesi, Á.; Szép, R. Major air pollutants seasonal variation analysis and long-range transport of PM₁₀ in an urban environment with specific climate condition in Transylvania (Romania). *Environ. Sci. Pollut. Res.* **2020**, *27*, 38181–38199. [CrossRef]
13. Shikhovtsev, M.Y.; Obolkin, V.; Khodzher, T.; Molozhnikova, Y.V. Variability of the ground concentration of particulate matter PM₁–PM₁₀ in the air basin of the Southern Baikal Region. *Atmos. Ocean. Opt.* **2023**, *36*, 655–662. [CrossRef]
14. Li, B.; Rodell, M. Spatial variability and its scale dependency of observed and modeled soil moisture over different climate regions. *Hydrol. Earth Syst. Sci.* **2013**, *17*, 1177–1188. [CrossRef]
15. Rojas, I.; Valenzuela, O.; Rojas, F.; Guillen, A.; Herrera, L.; Pomares, H.; Marquez, L.; Pasadas, M. Soft-computing techniques and ARMA model for time series prediction. *Neurocomputing* **2008**, *71*, 519–537. [CrossRef]
16. Aguilar, S.; Castro Souza, R.; Pessanha, J.F.; Cyrino Oliveira, F.L. Hybrid methodology for modeling short-term wind power generation using conditional Kernel density estimation and singular spectrum analysis. *Dyna* **2017**, *84*, 145–154. [CrossRef]
17. López-Gonzales, J.L.; Salas, R.; Velandia, D.; Canas Rodrigues, P. Air quality prediction based on singular spectrum analysis and artificial neural networks. *Entropy* **2024**, *26*, 1062. [CrossRef]

18. Espinosa, F.; Bartolomé, A.B.; Hernández, P.V.; Rodríguez-Sánchez, M. Contribution of singular spectral analysis to forecasting and anomalies detection of indoors air quality. *Sensors* **2022**, *22*, 3054. [\[CrossRef\]](#)
19. Espinoza Guillen, J.A. Evaluación Espacial y Temporal del Material Particulado PM₁₀ y PM_{2.5} en Lima Metropolitana para el Periodo 2015–2017. Bachelor's Thesis, Universidad Nacional Agraria La Molina, Lima, Peru, 2018.
20. Silva, J.; Rojas, J.; Norabuena, M.; Molina, C.; Toro, R.A.; Leiva-Guzmán, M.A. Particulate matter levels in a South American megacity: The metropolitan area of Lima-Callao, Peru. *Environ. Monit. Assess.* **2017**, *189*, 635. [\[CrossRef\]](#)
21. Aceves-Fernandez, M.A.; Pedraza-Ortega, J.C.; Sotomayor-Olmedo, A.; Ramos-Arreguín, J.M.; Vargas-Soto, J.E.; Tovar-Arriaga, S. Analysis of key features of non-linear behaviour using recurrence quantification. Case study: Urban Airborne pollution at Mexico city. *Environ. Model. Assess.* **2014**, *19*, 139–152. [\[CrossRef\]](#)
22. Box, G.; Jenkins, G. *Time Series Analysis: Forecasting and Control*; Holden-Day series in time series analysis and digital processing; Holden-Day: Clarendon, Australia, 1970.
23. Hyndman, R.J.; Khandakar, Y. Automatic time series forecasting: The forecast package for R. *J. Stat. Softw.* **2008**, *27*, 1–22. [\[CrossRef\]](#)
24. Brown, R.G. *Statistical Forecasting for Inventory Control*; McGraw-Hill: New York, NY, USA, 1959.
25. Holt, C.C. Forecasting seasonals and trends by exponentially weighted moving averages. *Int. J. Forecast.* **2004**, *20*, 5–10. [\[CrossRef\]](#)
26. Winters, P.R. Forecasting sales by exponentially weighted moving averages. *Manag. Sci.* **1960**, *6*, 324–342. [\[CrossRef\]](#)
27. Hyndman, R.; Athanasopoulos, G. *Forecasting: Principles and Practice*, 2nd ed.; OTexts: Melbourne, Australia, 2018.
28. Hyndman, R.; Koehler, A.; Ord, K.; Snyder, R. *Forecasting with Exponential Smoothing. The State Space Approach*; Springer: Berlin/Heidelberg, Germany, 2008. [\[CrossRef\]](#)
29. Zhang, G.P.; Patuwo, B.E.; Hu, M.Y. Forecasting with artificial neural networks: The state of the art. *Int. J. Forecast.* **1998**, *14*, 35–62. [\[CrossRef\]](#)
30. Crone, S.F.; Kourentzes, N. Feature selection for time series prediction—A combined filter and wrapper approach for neural networks. *Neurocomputing* **2010**, *73*, 1923–1936. [\[CrossRef\]](#)
31. Goodfellow, I.; Bengio, Y.; Courville, A. *Deep Learning*; MIT Press: Cambridge, MA, USA, 2016.
32. Glorot, X.; Bordes, A.; Bengio, Y. Deep sparse rectifier neural networks. In Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, Fort Lauderdale, FL, USA, 11–13 April 2011; Volume 15, pp. 315–323.
33. Hornik, K.; Stinchcombe, M.; White, H. Multilayer feedforward networks are universal approximators. *Neural Netw.* **1989**, *2*, 359–366. [\[CrossRef\]](#)
34. Hochreiter, S.; Schmidhuber, J. Long short-term memory. *Neural Comput.* **1997**, *9*, 1735–1780. [\[CrossRef\]](#)
35. Gers, F.A.; Schmidhuber, J.; Cummins, F. Learning to forget: Continual prediction with LSTM. *Neural Comput.* **2002**, *12*, 2451–2471. [\[CrossRef\]](#) [\[PubMed\]](#)
36. Greff, K.; Srivastava, R.K.; Koutník, J.; Steunebrink, B.R.; Schmidhuber, J. LSTM: A search space odyssey. *IEEE Trans. Neural Networks Learn. Syst.* **2017**, *28*, 2222–2232. [\[CrossRef\]](#)
37. West, M.; Harrison, J. *Bayesian Forecasting and Dynamic Models*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 1997.
38. Petris, G. An R Package for Dynamic Linear Models. *J. Stat. Softw.* **2010**, *36*, 1–16. [\[CrossRef\]](#)
39. Gonzales, S.M.; Iftikhar, H.; López-Gonzales, J.L. Analysis and forecasting of electricity prices using an improved time series ensemble approach: An application to the Peruvian electricity market. *Aims Math* **2024**, *9*, 21952–21971. [\[CrossRef\]](#)
40. Iftikhar, H.; Gonzales, S.M.; Zywiłk, J.; López-Gonzales, J.L. Electricity demand forecasting using a novel time series ensemble technique. *IEEE Access* **2024**, *12*, 88963–88975. [\[CrossRef\]](#)
41. da Silva, K.L.S.; López-Gonzales, J.L.; Turpo-Chaparro, J.E.; Tocto-Cano, E.; Rodrigues, P.C. Spatio-temporal visualization and forecasting of PM₁₀ in the Brazilian state of Minas Gerais. *Sci. Rep.* **2023**, *13*, 3269. [\[CrossRef\]](#)
42. Cruz, A.R.H.D.L.; Ayuque, R.F.O.; Cruz, R.W.H.D.L.; Lopez-Gonzales, J.L.; Gioda, A. Air quality biomonitoring of trace elements in the metropolitan area of Huancayo, Peru using transplanted Tillandsia capillaris as a biomonitor. *An. Acad. Bras. Ciências* **2020**, *92*, e20180813. [\[CrossRef\]](#) [\[PubMed\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.